Problem 1. Find the derivative

$$\frac{d}{dx}\left\{\int_{\sqrt{x}}^{\cos x}\frac{\sqrt{t^3+1}}{t+7}\,dt\right\}.$$

Sln.

$$\frac{d}{dx}\left\{\int_{f(x)}^{h(x)} g(t)\,dt\right\} = g(h(x))\cdot h'(x) - g(f(x))\cdot f'(x).$$

$$\int_{f(x)}^{h(x)} g(t)\,dt = \int_{f(x)}^{0} g(t)\,dt + \int_{0}^{h(x)} g(t)\,dt$$

$$= \int_{0}^{h(x)} g(t)\,dt - \int_{0}^{f(x)} g(t)\,dt.$$

$$\frac{d}{dx}\left\{\int_{\sqrt{x}}^{\cos x}\frac{\sqrt{t^3+1}}{t+7}\,dt\right\} = \frac{\sqrt{\cos^3 x+1}}{\cos x+7}\cdot\frac{d}{dx}(\cos x) - \frac{\sqrt{x^{3/2}+1}}{\sqrt{x}+7}\cdot\frac{d}{dx}(\sqrt{x}).$$

$$= -\frac{\sin x\sqrt{\cos^3 x+1}}{\cos x+7} - \frac{\sqrt{x^{3/2}+1}}{2\sqrt{x}(\sqrt{x}+7)}.$$

Problem 2. A particle moves along a straight line so that the velocity at time t is $v(t)=2t-6$. Find the total distance traveled during the time period $0\le t\le 4$.

Sln. Notice that $v(t)=2t-6$

$v(t)\le 0$ for $0\le t\le 3$

$v(t)\ge 0$ for $3\le t\le 4$.

$$\text{Total distance} = \int_0^4 |v(t)|\,dt = \int_0^3 |v(t)|\,dt + \int_3^4 |v(t)|\,dt$$

$$= \int_0^3 -v(t)\,dt + \int_3^4 v(t)\,dt$$

$$= -\int_0^3 (2t-6)\,dt + \int_3^4 (2t-6)\,dt$$

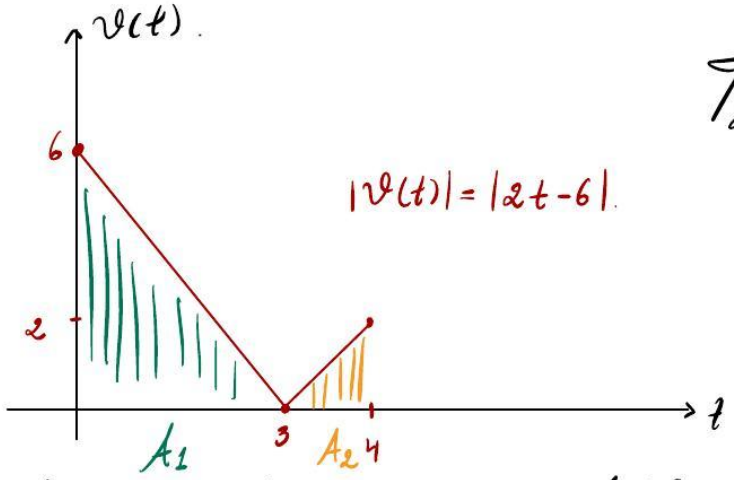
$$= -(t^2-6t)\Big|_0^3 + (t^2-6t)\Big|_3^4$$

$= -(9-18) + (16-24-(9-18))$

$= -2(-9) - 8$

$= 10.$

2nd Way:



$|v(t)| = |2t-6|.$

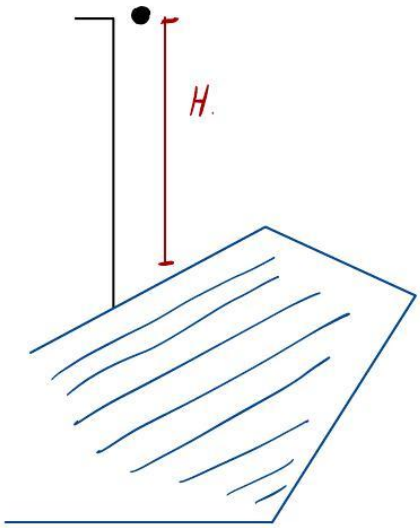
$$\text{Total} = \int_0^4 |v(t)|\,dt = A_1 + A_2.$$

$$= \frac{1}{2}(3)(6) + \frac{1}{2}(1)(2)$$

$$= 9 + 1 = 10.$$

Problem 3. A rock is dropped from a cliff and hits the surface of the lake below 3 seconds later. Find the distance between the rock and the surface of the lake as a function of the time starting from the moment it was dropped.

Note: Assume that the acceleration of gravity is 10 m/s^2 and ignore the drag.



$a(t) = -10\ m/s^2.$

$v(t) = \int a(t)\,dt = \int (-10)\,dt = -10t + C.$

at $t = 0s$ then $v(0) = 0\ m/s.$

$0 = -10(0) + C$ so, $C = 0\ m/s.$

$[0] = [-10] \times [0] + [C].$

$m/s \quad m/s^2 \cdot s \quad m/s$

$$v(t) = -10t$$

$$H(t) = \int v(t)\,dt = \int -10t\,dt = -5t^2 + D.$$

when $t = 3s$ then $H(3s) = 0$. So,

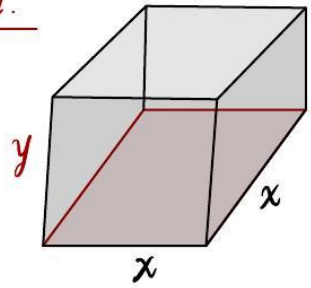
$$0 = -5(3)^2 + \mathcal{Q}$$

$$\mathcal{Q} = 45.$$

So, $\boxed{H(t) = -5t^2 + 45.}$

Problem 4. We are planning to make a box with a open top and square base. It has to have the volume of $6\,ft^3$. The material for the base costs \$3.00 per square foot and the material for the sides costs \$2.00 per square foot. Find the dimentions of such box that result in the smallest possible total cost of material. justify.

Sln.



$$\mathcal{V} = x \cdot x \cdot y = x^2 \cdot y = 6\,ft^3.$$

$$C = 3 \cdot x^2 + 2 \cdot (4xy).$$

cost base

from $x^2 \cdot y = 6$ $\quad y = \frac{6}{x^2}$

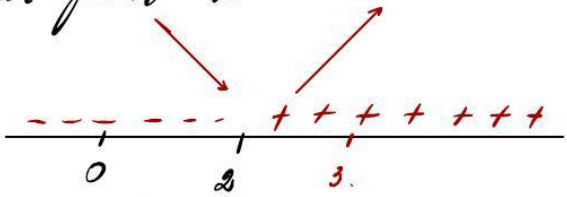
$$C(x) = 3x^2 + 8x \cdot \left(\frac{6}{x^2}\right) = 3x^2 + \frac{48}{x}.$$

$$C'(x) = 6x - \frac{48}{x^2} = \frac{6x^3 - 48}{x^2} \Rightarrow \boxed{x = 0} \text{ or } x = \sqrt[3]{\frac{48}{6}} = 2.$$

x

The only critical point $x = 2$.

$C'(x) \geq 0$



By the 1st derivative test at $x = 2$ we have a minimum.

$$y = \frac{6}{x^2} = \frac{6}{4} = \frac{3}{2}.$$

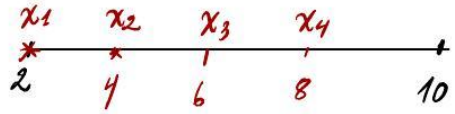
$\boxed{2\,ft \times 2\,ft \times \frac{3}{2}\,ft}$ dimentions.

Problem 5. a). Calculate the left-point approximation L_n to integral.

$$\int_2^{10} \frac{1}{x}\,dx.$$

b). Draw the corresponding rectangles together with grah of $f(x) = \frac{1}{x}$.

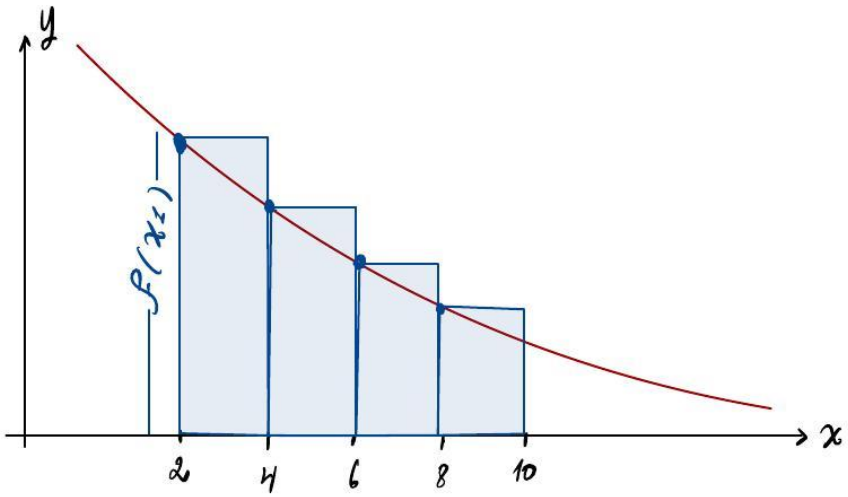
Sln a)



$$\Delta x = \frac{10-2}{4} = 2.$$

$$L_4 = f(x_1)\Delta x + f(x_2)\Delta x + f(x_3)\Delta x + f(x_4)\Delta x$$
$$= 2\left(f(x_1) + f(x_2) + f(x_3) + f(x_4)\right)$$
$$= 2\left(\frac{1}{2} + \frac{1}{4} + \frac{1}{6} + \frac{1}{8}\right) = \frac{25}{12}.$$

b).



c). Is L_4 greater than the actual integral or less than it?

Specify and explain, or explain why this cannot be determinated.

Sln c). L_4 is greater than $\int_2^{10} \frac{1}{x}\,dx$, b/c the total area of rectangles from part b). is greater than the area below the curve $y = f(x)$.

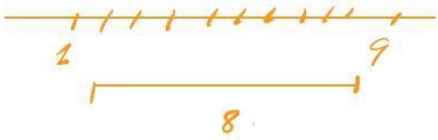
Problem 6. Express the following limit as an integral.

$$\lim_{n\to\infty} \sum_{i=1}^{n} \left\{ \sqrt{\left(1+\frac{8i}{n}\right)^2 + 3\left(1+\frac{8i}{n}\right)} \left(\frac{8}{n}\right) \right\}.$$

Sln. $$\int_a^b f(x)\,dx = \lim_{n\to\infty} \sum_{i=1}^{n} f(x_i)\Delta x \qquad \Delta x = \frac{b-a}{n}.$$

$$x_i = a + i\Delta x.$$

$$\Delta x = \frac{8}{n} \qquad x_i = 1 + \frac{8}{n} i \qquad a = 1, \quad b = 9$$



$$\int_1^9 \sqrt{x^2 + 3x}\, dx. \qquad f(x) = \sqrt{x^2 + 3x}$$